[C PROGRAMMING ASSIGNMENT]

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COURSE - B.TECH(ELECTRICAL ENGINEERING)

SUBJECT - PROGRAMMING FUNDAMENTALS FOR ELECTRONICS ENGINEERING

ADMISSION NUMBER - 25SECE1050025

C PROGRAMMING
1.MATRIX MULTIPLICATION
#CODE
```
#include<stdio.h>

int main() {
        int r1, c1, r2, c2, i, j, k;

    printf("Enter rows and cols for matrix A: ");
    scanf("%d %d", &r1, &c1);

        printf("Enter rows and cols for matrix B: ");
    scanf("%d %d", &r2, &c2);

        if(c1 != r2) {
        printf("Error: Matrix multiplication not possible\n");
        return 0;
        }

        int a[r1][c1], b[r2][c2], result[r1][c2];

    printf("Enter elements of matrix A:\n");
        for(i = 0; i < r1; i++) {
        for(j = 0; j < c1; j++) {
            scanf("%d", &a[i][j]);
        }
        }

    printf("Enter elements of matrix B:\n");
        for(i = 0; i < r2; i++) {
        for(j = 0; j < c2; j++) {
```

```c
            scanf("%d", &b[i][j]);
        }
        }
        for(i = 0; i < r1; i++) {
        for(j = 0; j < c2; j++) {
            result[i][j] = 0;
        }
        }
        for(i = 0; i < r1; i++) {
        for(j = 0; j < c2; j++) {
            for(k = 0; k < c1; k++) {
                result[i][j] += a[i][k] * b[k][j];
        }
        }
        }

    printf("Result matrix:\n");

        for(i = 0; i < r1; i++) {
        for(j = 0; j < c2; j++) {
            printf("%d ", result[i][j]);
        }
        printf("\n");
        }

        return 0;
}
```

#output#

Output Clear

```
Enter rows and cols for matrix A: 2
3
Enter rows and cols for matrix B: 3
4
Enter elements of matrix A:
2
3
3
4
5
6
Enter elements of matrix B:
3
4
6
7
8
8
9
7
6
5
4
3
Result matrix:
48 47 51 44
88 86 93 81


=== Code Execution Successful ===
```

2.String palindrome without library functions

#CODE

```
#include <stdio.h>

int main() {
	char str[100];
	int length = 0, i, isPalindrome = 1;
printf("Enter a string: ");
	scanf("%s", str);

	while (str[length] != '\0') {
	length++;
	}

	for (i = 0; i < length / 2; i++) {
	// Compare character at index i with character at
symmetric position from end
	if (str[i] != str[length - 1 - i]) {
	isPalindrome = 0; // Not a palindrome
	break;
	}
	}

	if (isPalindrome) {
	printf("%s is a palindrome.\n", str);
	} else {
	printf("%s is not a palindrome.\n", str);
	}
	return 0;
```

```
}
```

#OUTPUT#

Output Clear

```
Enter a string: abhijeet
abhijeet is not a palindrome.


=== Code Execution Successful ===
```

Output

Clear

```
Enter a string: nitin
nitin is a palindrome.


=== Code Execution Successful ===
```

3. File handling - Count words , characters ,lines

#CODE

```
#include <stdio.h>
#include <ctype.h>

int main() {
    FILE *file;
    char filename[100];
    int ch;
    int characters = 0, words = 0, lines = 0;
    int in_word =

    printf("Enter the filename: ");
    scanf("%s", filename);

    file = fopen(filename, "r");

    if (file == NULL) {
    printf("Could not open file %s\n", filename);
    return 1;
    }

    while ((ch = fgetc(file)) != EOF) {
        characters++;


        if (ch == '\n') {
            lines++;
        }
```

```
    if (isspace(ch)) {
    in_word = 0;
    } else if (in_word == 0) {
    in_word = 1;
    words++;
    }
    }


    fclose(file);
    printf("\n--- File Statistics ---\n");
    printf("Characters: %d\n", characters);
    printf("Words:      %d\n", words);
    printf("Lines:      %d\n", lines);

    return 0;
}
```

#output#

Output

Clear

```
Enter the filename: abhijeet.txt
Could not open file abhijeet.txt


=== Code Exited With Errors ===
```

4. structure - student records

#code

```
#include <stdio.h>


struct Student {
    char name[50];
    int roll;
    float marks;
};

int main() {
    int n, i, topperIndex = 0;

    printf("Enter number of students: ");
    scanf("%d", &n);

    struct Student s[n];

    for(i = 0; i < n; i++) {
        printf("\nEnter details for Student %d:\n", i + 1);
        printf("Name: ");
        scanf("%s", s[i].name);
        printf("Roll Number: ");
        scanf("%d", &s[i].roll);
        printf("Marks: ");
        scanf("%f", &s[i].marks);


        if (s[i].marks > s[topperIndex].marks) {
            topperIndex = i;
```

```
        }
    }

    printf("\n--- All Student Records ---\n");
    for(i = 0; i < n; i++) {
        printf("Roll: %d | Name: %s | Marks: %.2f\n", s[i].roll,
s[i].name, s[i].marks);
    }

    printf("\n--- TOPPER DETAILS ---\n");
    printf("Name: %s\n", s[topperIndex].name);
    printf("Roll No: %d\n", s[topperIndex].roll);
    printf("Highest Marks: %.2f\n", s[topperIndex].marks);

    return 0;
}
```

#OUTPUT#

Output
Clear

```
Enter number of students: 5

Enter details for Student 1:
Name: Abhijeet
Roll Number: 1
Marks: 80

Enter details for Student 2:
Name: Arman
Roll Number: 2
Marks: 90

Enter details for Student 3:
Name: Ajay
Roll Number: 3
Marks: 78

Enter details for Student 4:
Name: Bhuvan
Roll Number: 4
Marks: 75

Enter details for Student 5:
Name: Daksh
Roll Number: 5
Marks: 87

--- All Student Records ---
Roll: 1 | Name: Abhijeet | Marks: 80.00
Roll: 2 | Name: Arman | Marks: 90.00
Roll: 3 | Name: Ajay | Marks: 78.00
Roll: 4 | Name: Bhuvan | Marks: 75.00
Roll: 5 | Name: Daksh | Marks: 87.00

--- TOPPER DETAILS ---
Name: Arman
Roll No: 2
Highest Marks: 90.00


=== Code Execution Successful ===
```

5.pointers - based array sum

#code

```
#include <stdio.h>
int main() {
    int length, sum = 0;
    printf("Enter the number of elements: ");
    scanf("%d", &length);

    int arr[length];
    printf("Enter %d elements:\n", length);
    for (int i = 0; i < length; i++) {
        scanf("%d", &arr[i]);
    }
    int *ptr = arr;
    for (int i = 0; i < length; i++) {
        sum += *(ptr + i);
    }
    printf("The sum of the array elements is: %d\n", sum);

    return 0;
}
```

#output#

Output   Clear

```
Enter the number of elements: 5
Enter 5 elements:
1
2
3
4
5
The sum of the array elements is: 15


=== Code Execution Successful ===
```

6.Sorting(without Built -in)

#code

```
#include <stdio.h>
int main() {
	int n, i, j, temp;
	printf("Enter number of elements: ");
	scanf("%d", &n);
	int arr[n];
	printf("Enter %d elements:\n", n);
	for(i = 0; i < n; i++) {
	scanf("%d", &arr[i]);
	}
	for(i = 0; i < n - 1; i++) {
	for(j = 0; j < n - i - 1; j++) {
	if(arr[j] > arr[j + 1]) {
		temp = arr[j];
		arr[j] = arr[j + 1];
		arr[j + 1] = temp;
	}
	}
	}
	printf("Sorted array in ascending order:\n");

	for(i = 0; i < n; i++) {
	printf("%d ", arr[i]);
	}

	return 0;
}
```

#OUTPUT#

Output Clear

```
Enter number of elements: 5
Enter 5 elements:
1 5 8 3 2
Sorted array in ascending order:
1 2 3 5 8

=== Code Execution Successful ===
```

7.Fibonacci using Recursion

#code

```
#include <stdio.h>
int fibonacci(int n) {
	if(n == 0)
	return 0;
	else if(n == 1)
	return 1;
	else
	return fibonacci(n - 1) + fibonacci(n - 2);
}
int main() {
	int n, i;
	printf("Enter number of terms: ");
	scanf("%d", &n);

	printf("Fibonacci Series:\n");
	for(i = 0; i < n; i++) {
	printf("%d ", fibonacci(i));
	}

	return 0;
}
```

#OUTPUT#

Output Clear

```
Enter number of terms: 10
Fibonacci Series:
0 1 1 2 3 5 8 13 21 34

=== Code Execution Successful ===
```

8. Find Second Largest Element

#code

```
#include <stdio.h>
int main() {
    int n, i;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter array elements:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int largest, secondLargest;
    if(arr[0] > arr[1]) {
        largest = arr[0];
        secondLargest = arr[1];
    } else {
        largest = arr[1];
        secondLargest = arr[0];
    }

    for(i = 2; i < n; i++) {
        if(arr[i] > largest) {
            secondLargest = largest;
            largest = arr[i];
        }
        else if(arr[i] > secondLargest && arr[i] != largest) {
            secondLargest = arr[i];
        }
```

```
    }

    printf("Second largest number = %d\n", secondLargest);

    return 0;
}
```

#OUTPUT#

Output Clear

```
Enter number of elements: 4
Enter array elements:
1
2
3
5
Second largest number = 3


=== Code Execution Successful ===
```

9. Menu-Driven Calculator

#code

```
#include <stdio.h>
float add(float a, float b);
float subtract(float a, float b);
float multiply(float a, float b);
float divide(float a, float b);
int main() {
    int choice;
    float num1, num2, result;
    while (1) {
        printf("\n===== Calculator Menu =====\n");
        printf("1. Add\n");
        printf("2. Subtract\n");
        printf("3. Multiply\n");
        printf("4. Divide\n");
        printf("5. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);
        if (choice == 5) {
            printf("Exiting program...\n");
            break;
        }

        printf("Enter two numbers: ");
        scanf("%f %f", &num1, &num2);

        switch (choice) {
            case 1:
```

```
            result = add(num1, num2);
            printf("Result = %.2f\n", result);
            break;

        case 2:
            result = subtract(num1, num2);
            printf("Result = %.2f\n", result);
            break;

        case 3:
            result = multiply(num1, num2);
            printf("Result = %.2f\n", result);
            break;

        case 4:
            result = divide(num1, num2);
            if (num2 != 0)
                printf("Result = %.2f\n", result);
            break;

        default:
            printf("Invalid choice!\n");
        }
    }

    return 0;
}

float add(float a, float b) {
    return a + b;
```

```
}

float subtract(float a, float b) {
    return a - b;
}

float multiply(float a, float b) {
    return a * b;
}

float divide(float a, float b) {
    if (b == 0) {
        printf("Division by zero is not allowed!\n");
        return 0;
    }
    return a / b;
}
```

#OUTPUT#

Output Clear

```
===== Calculator Menu =====
1. Add
2. Subtract
3. Multiply
4. Divide
5. Exit
Enter your choice: 1
Enter two numbers: 3
4
Result = 7.00

===== Calculator Menu =====
1. Add
2. Subtract
3. Multiply
4. Divide
5. Exit
Enter your choice: 5
Exiting program...
```